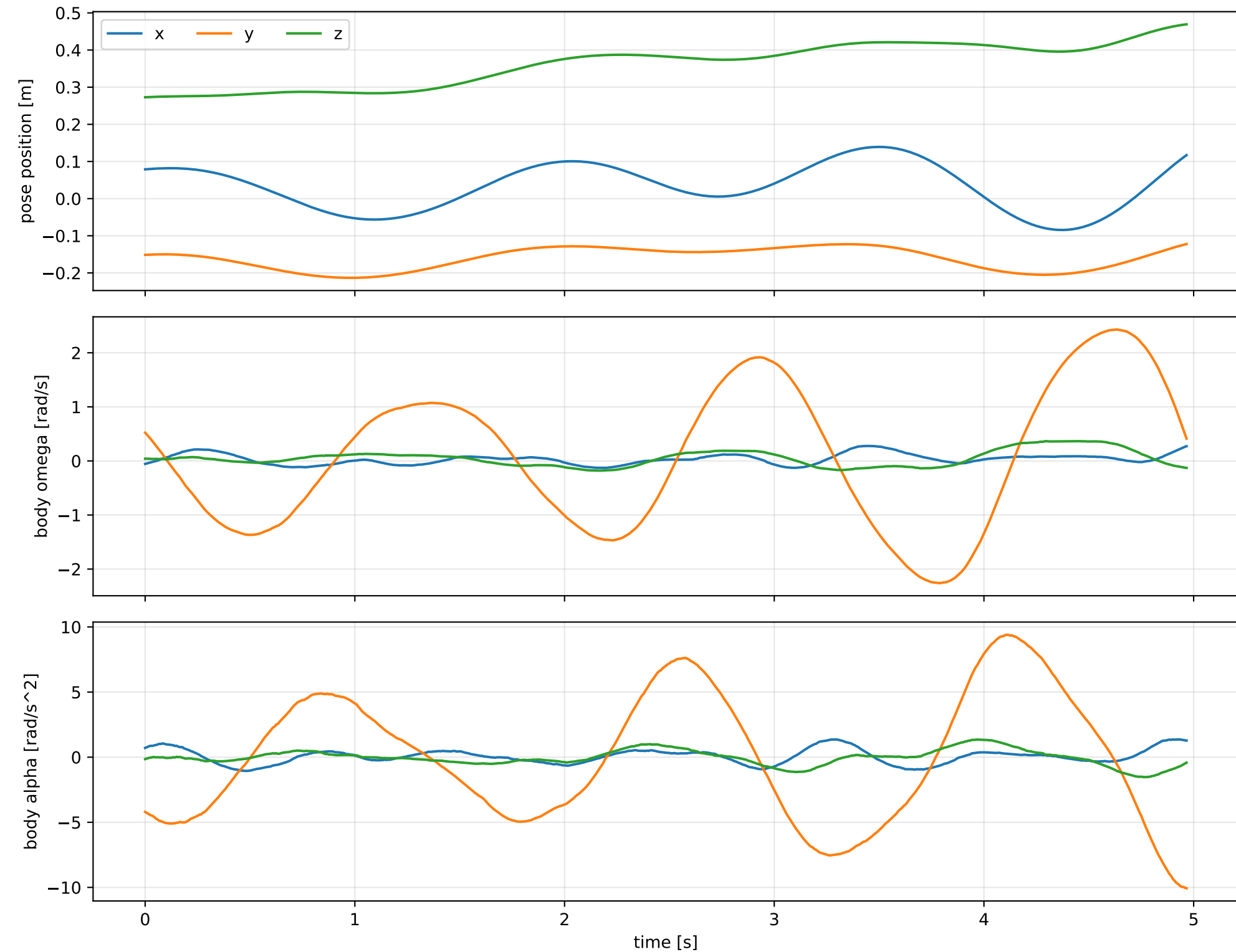
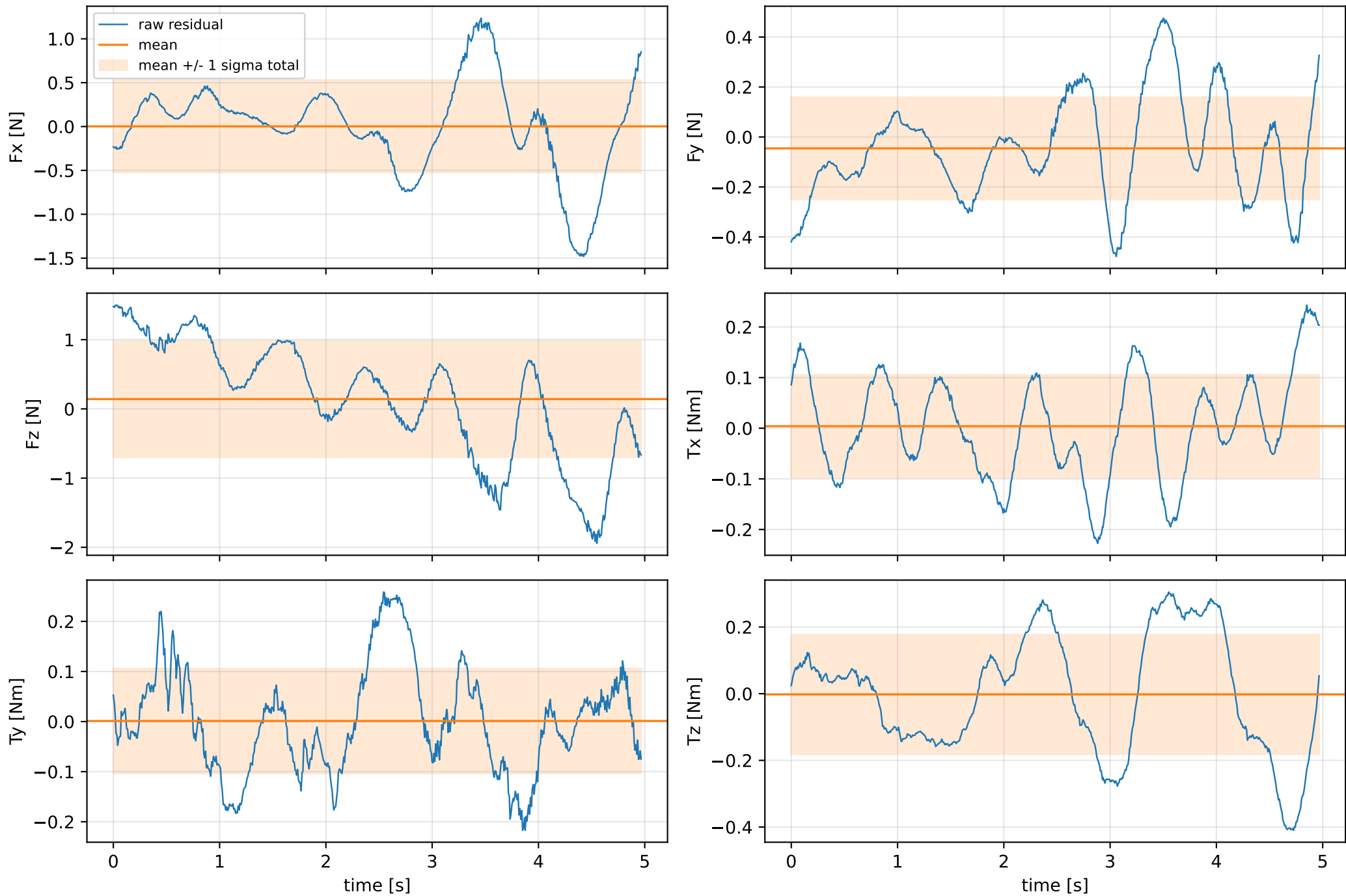


cog_prior_reseed_refined: SG trajectory and derived motion



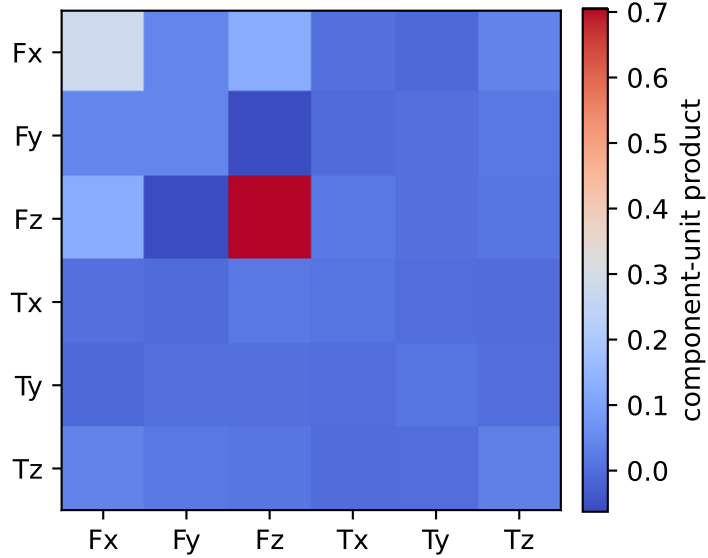
cog_prior_reseed_refined: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



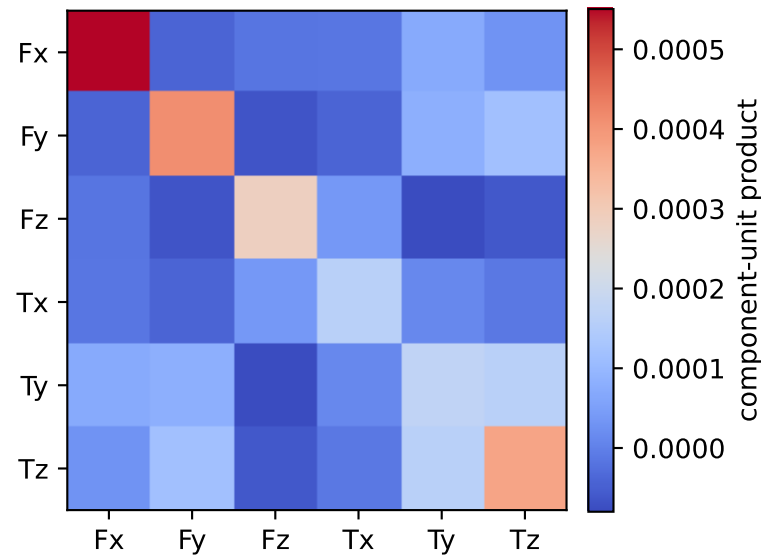
vector RMS about zero: force 1.0219 N, torque 0.2308 Nm; component std about mean: force [0.5268 0.2047 0.8393], torque [0.1014 0.1049 0.179]

cog_prior_reseed_refined: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.008 0.009 0.016 0.035 0.266 0.741]

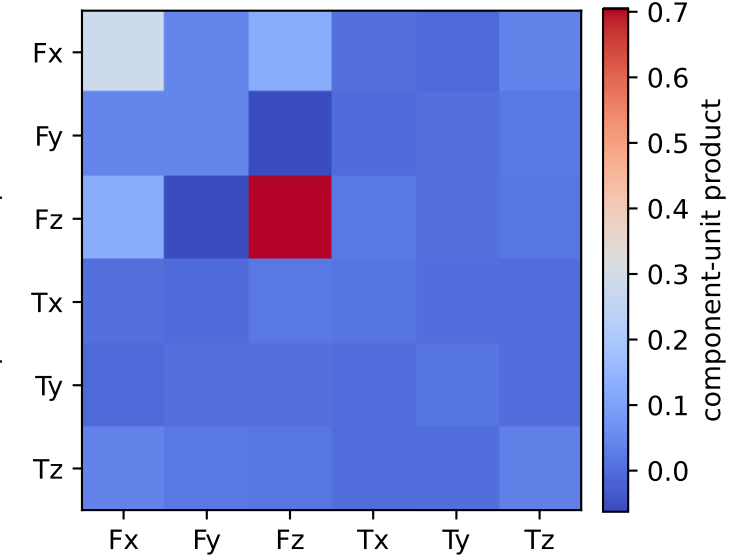
empirical total covariance



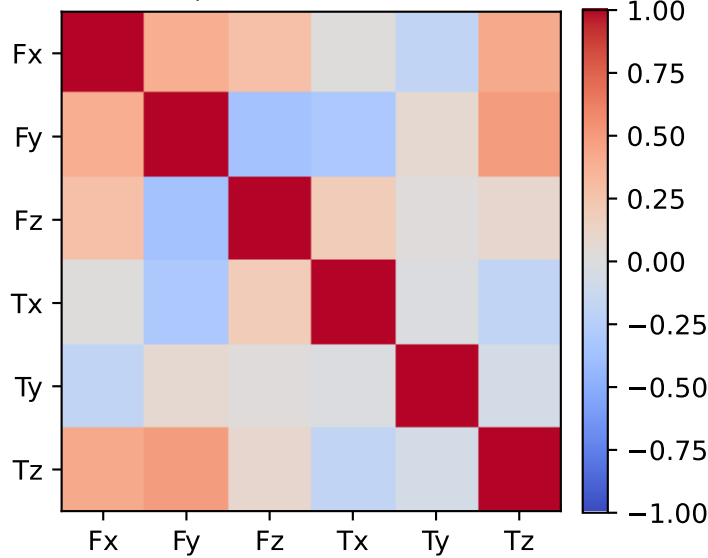
mean full-SG prediction covariance



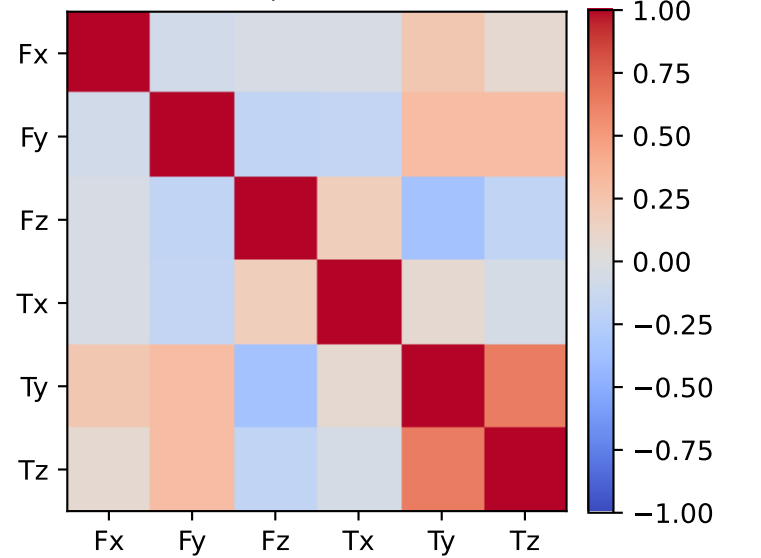
PSD excess model discrepancy covariance



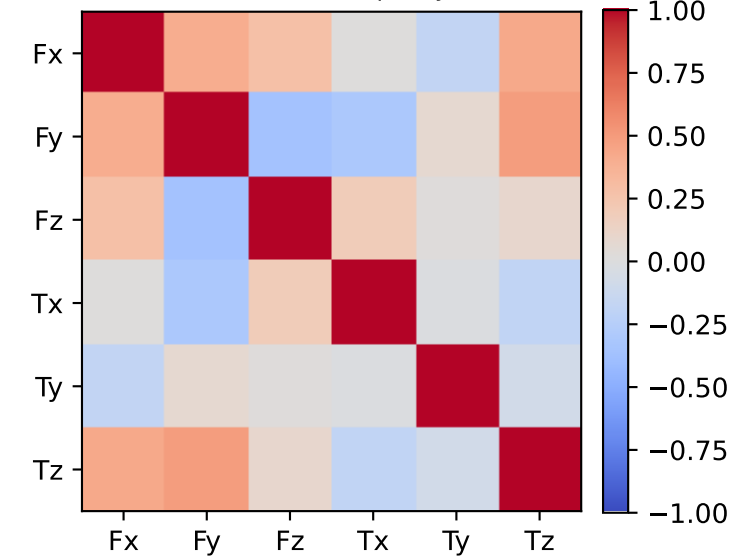
empirical total correlation



mean full-SG prediction correlation

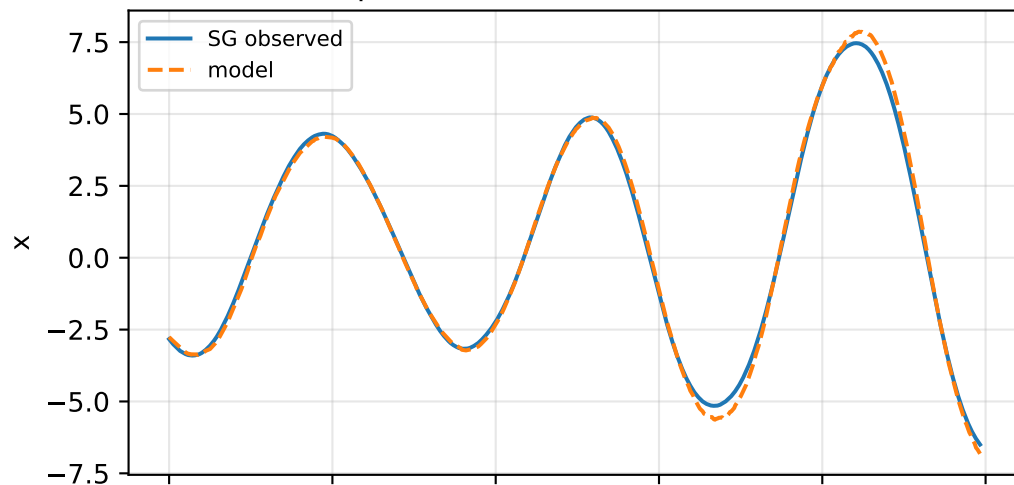


PSD excess model discrepancy correlation

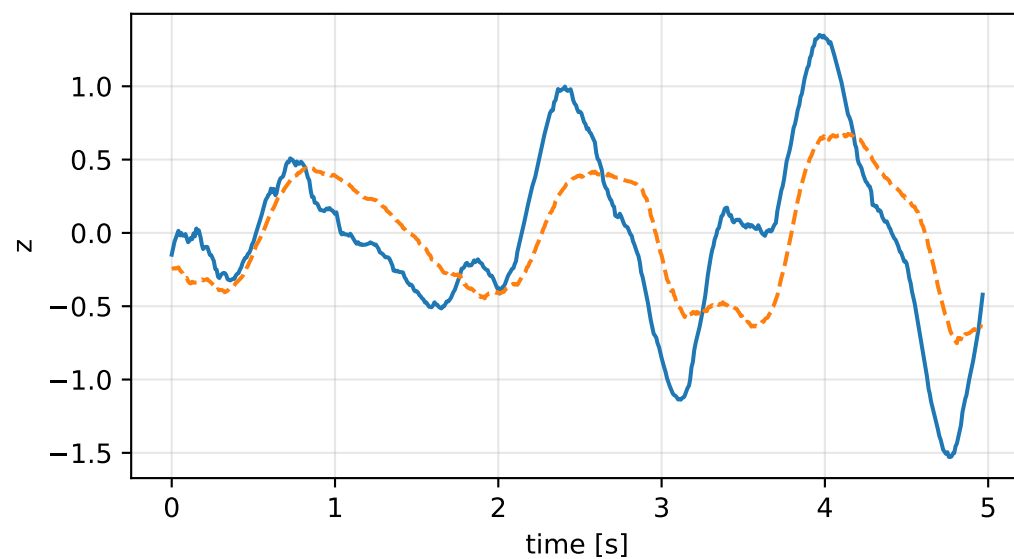
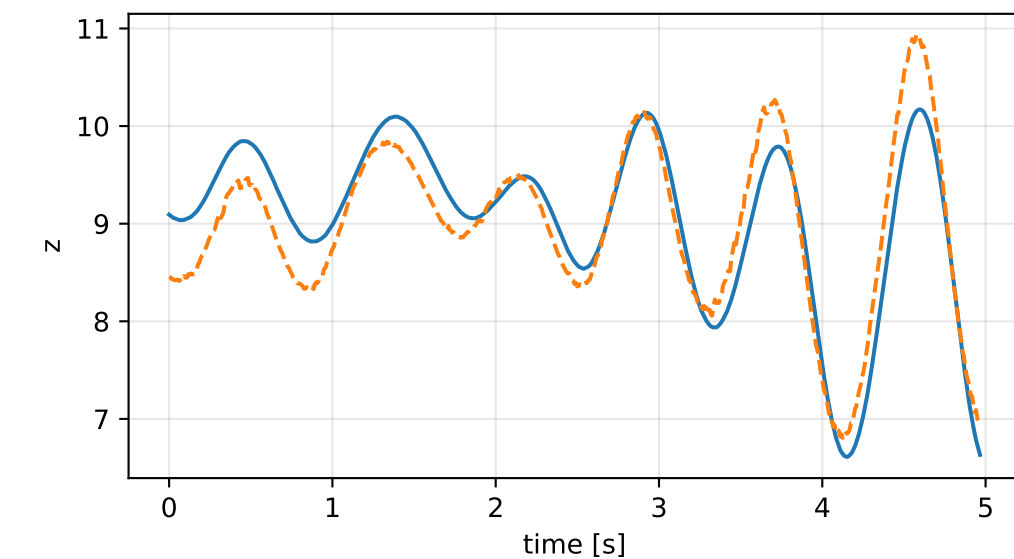
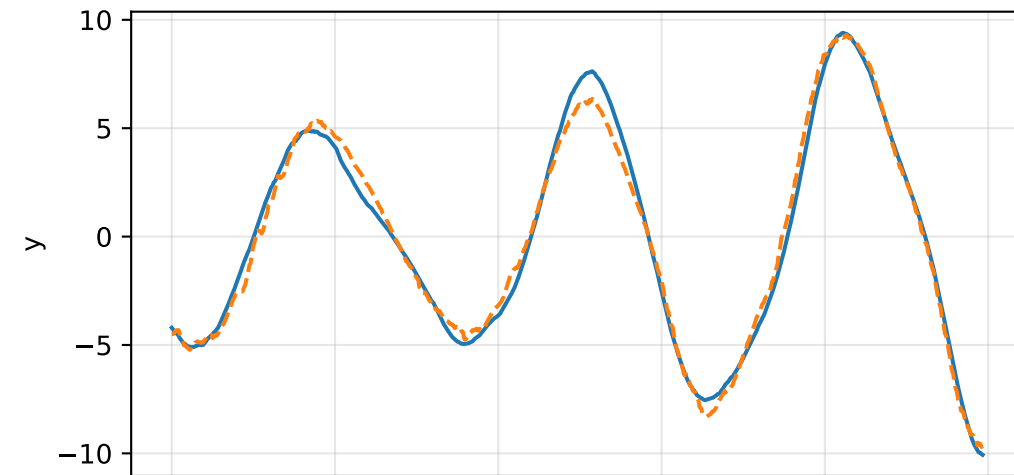
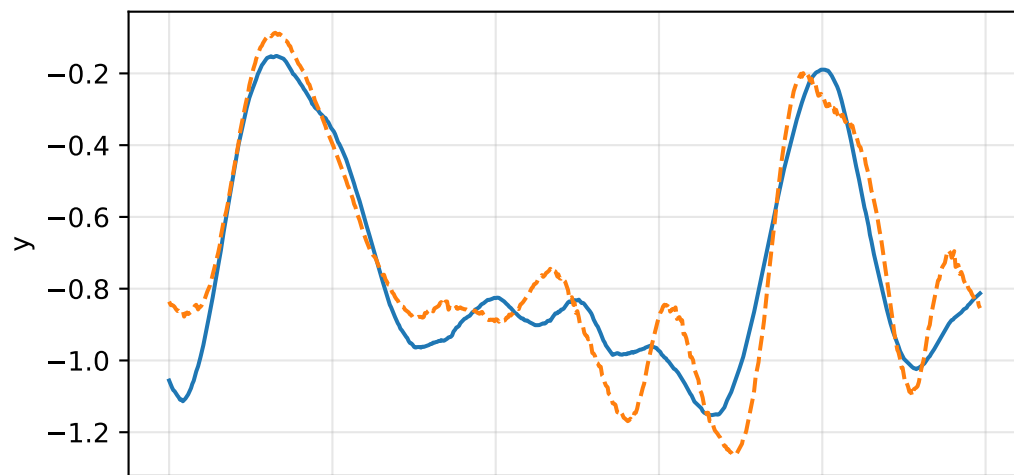
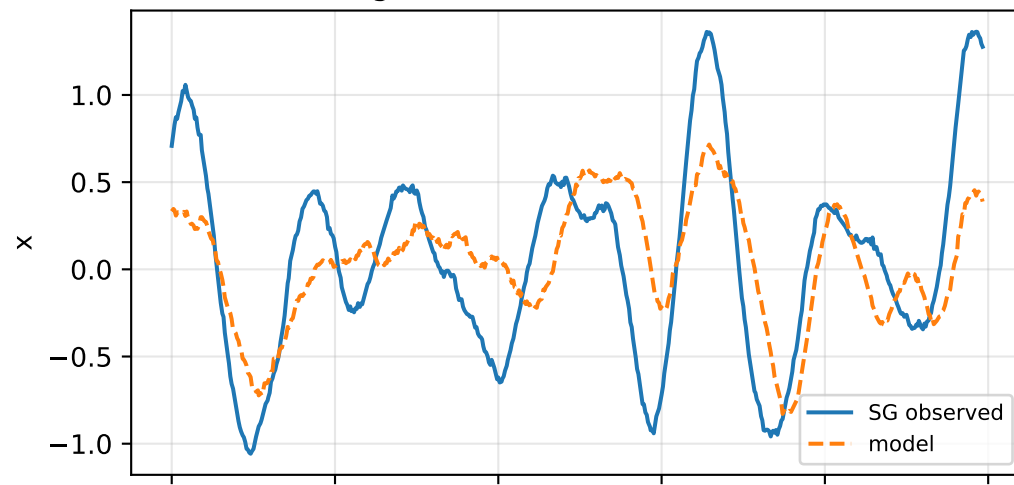


cog_prior_reseed_refined: acceleration residual objective

specific acceleration [m/s^2]

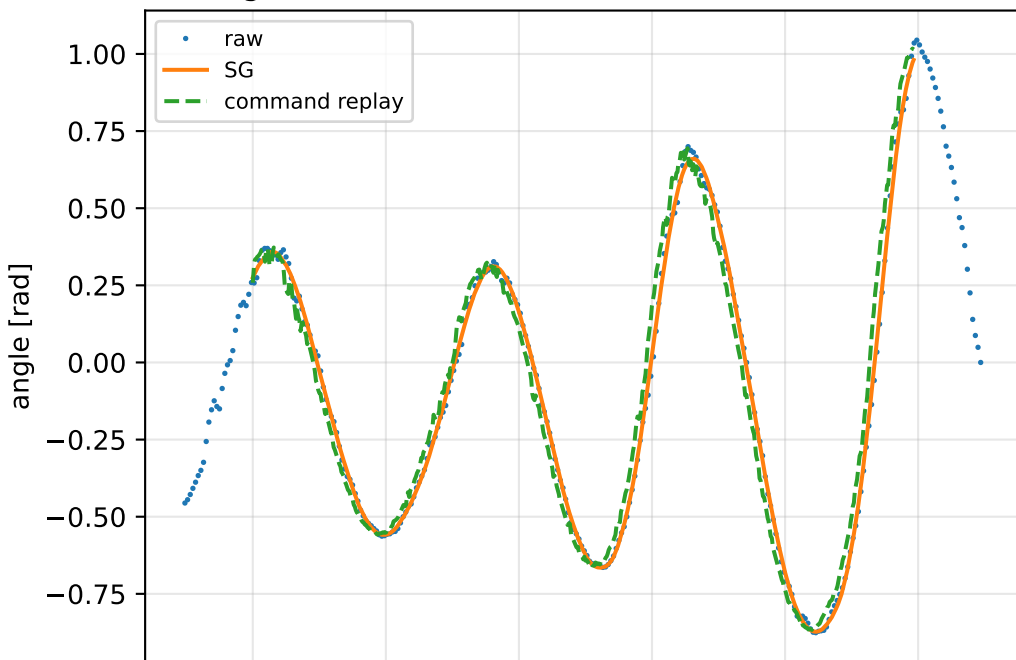


angular acceleration [rad/s^2]

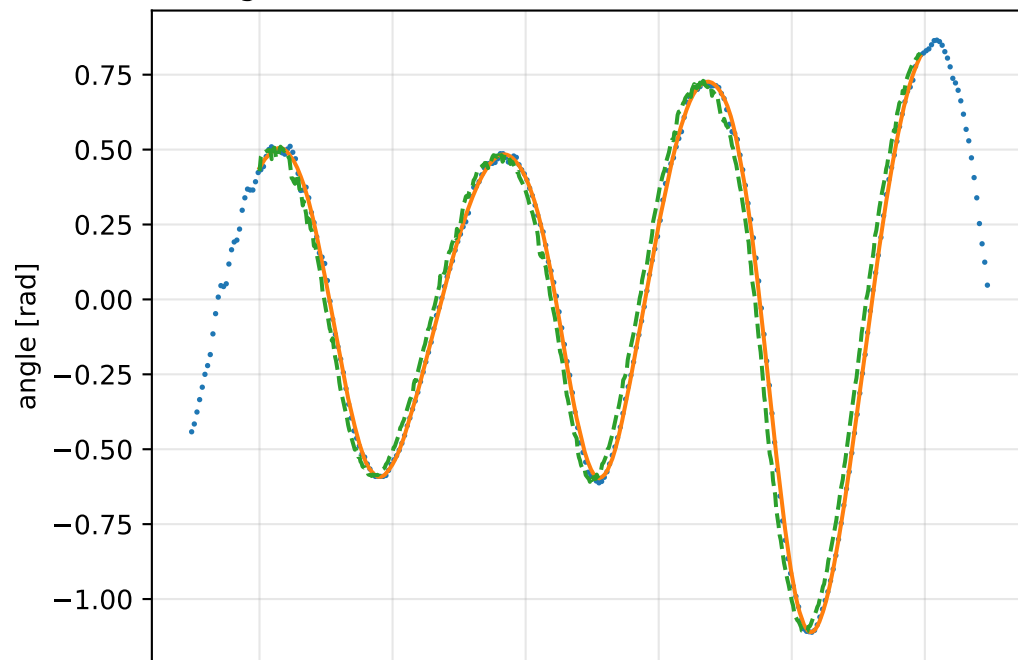


cog_prior_reseed_refined: gimbal measurement audit (objective=measured_sg)

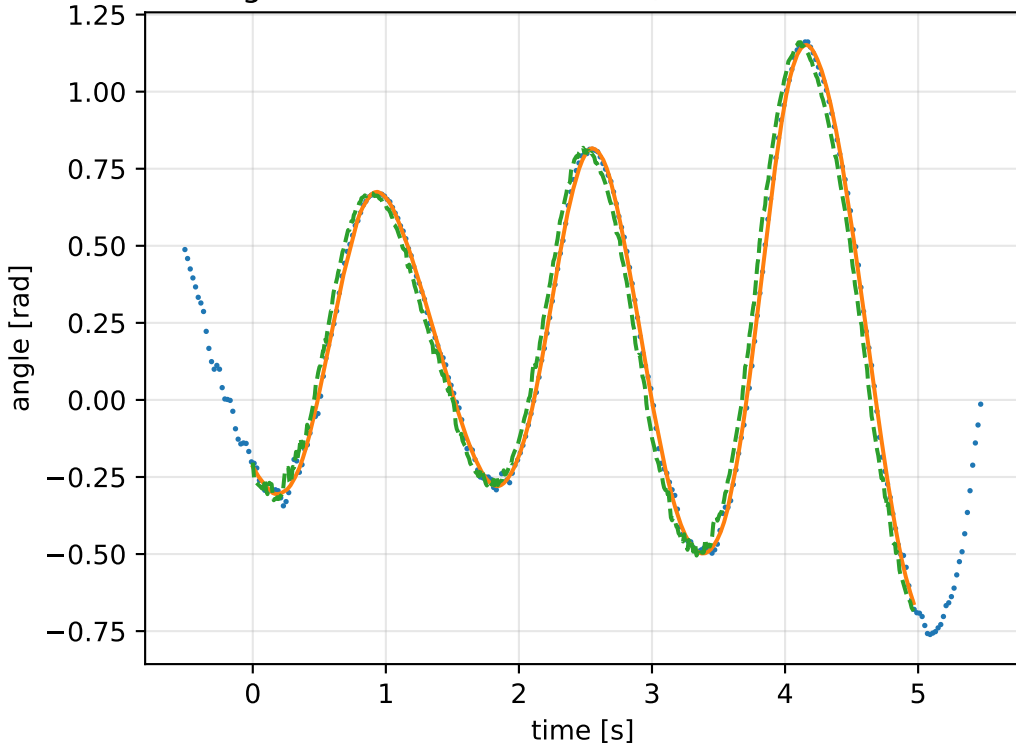
gimbal 1: RMSE=0.0759, max=0.2095 rad



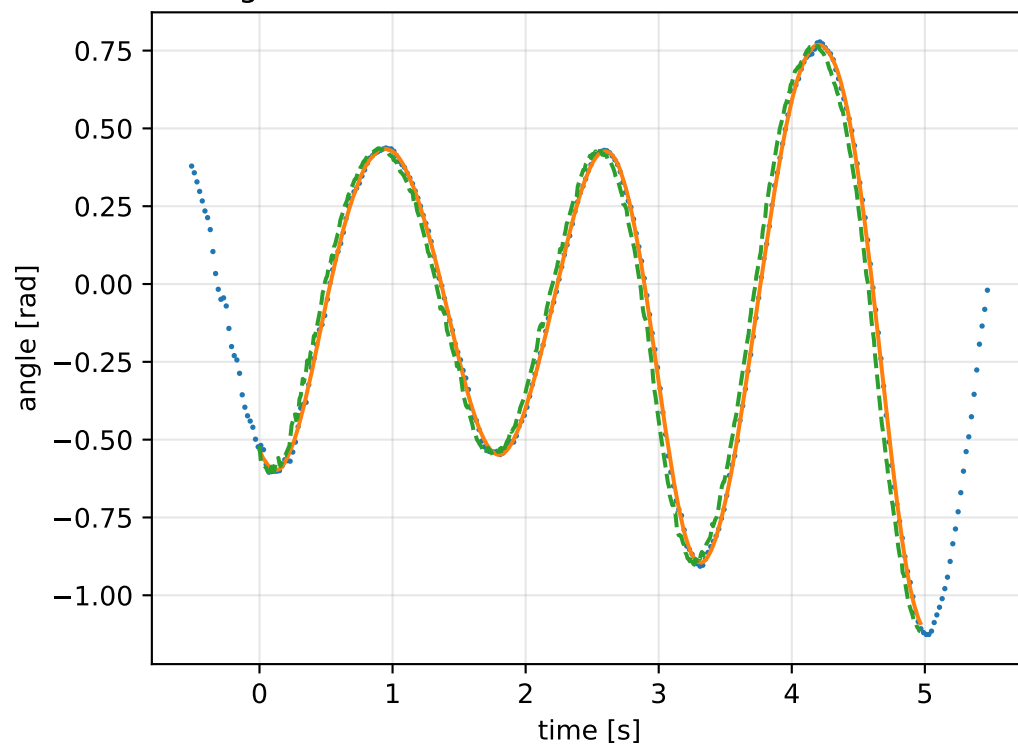
gimbal 2: RMSE=0.08099, max=0.2013 rad



gimbal 3: RMSE=0.07393, max=0.172 rad

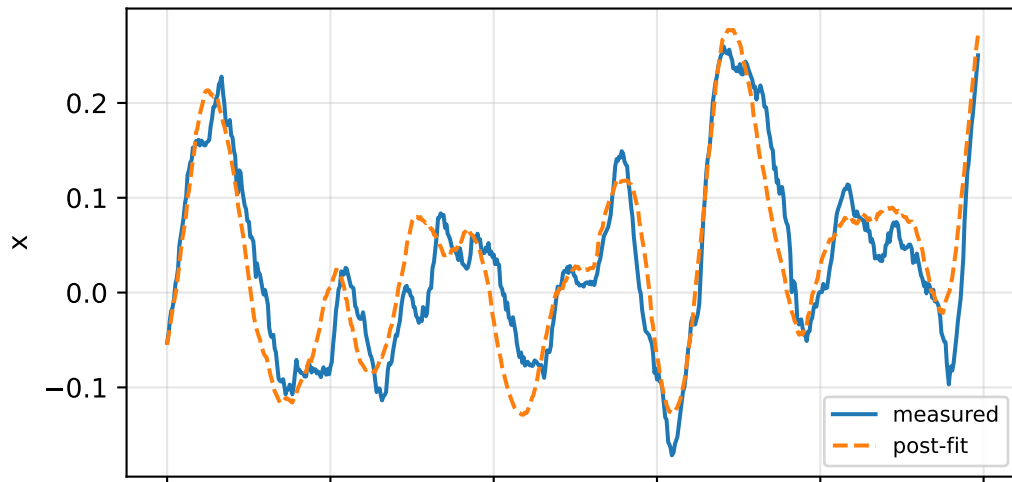


gimbal 4: RMSE=0.07131, max=0.1737 rad

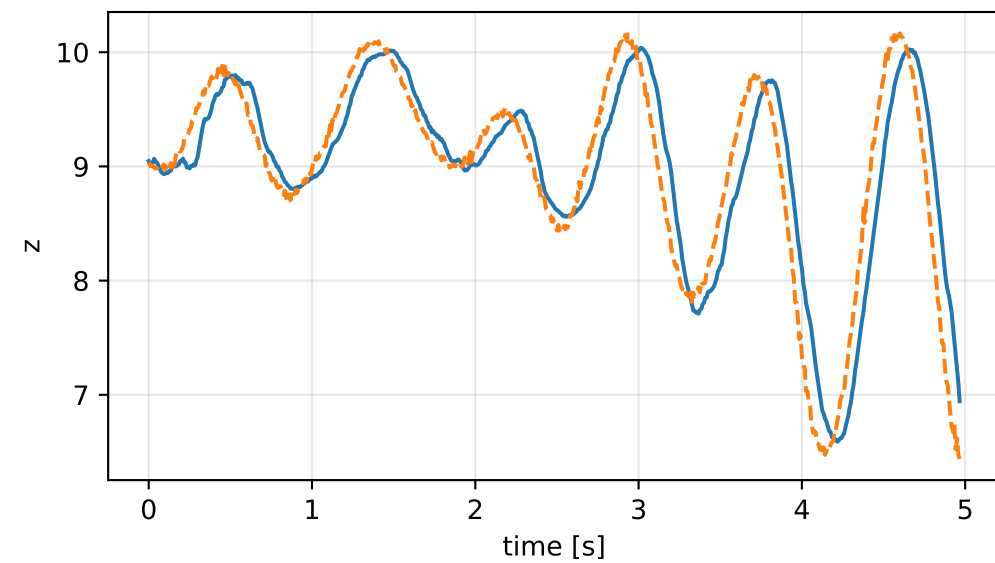
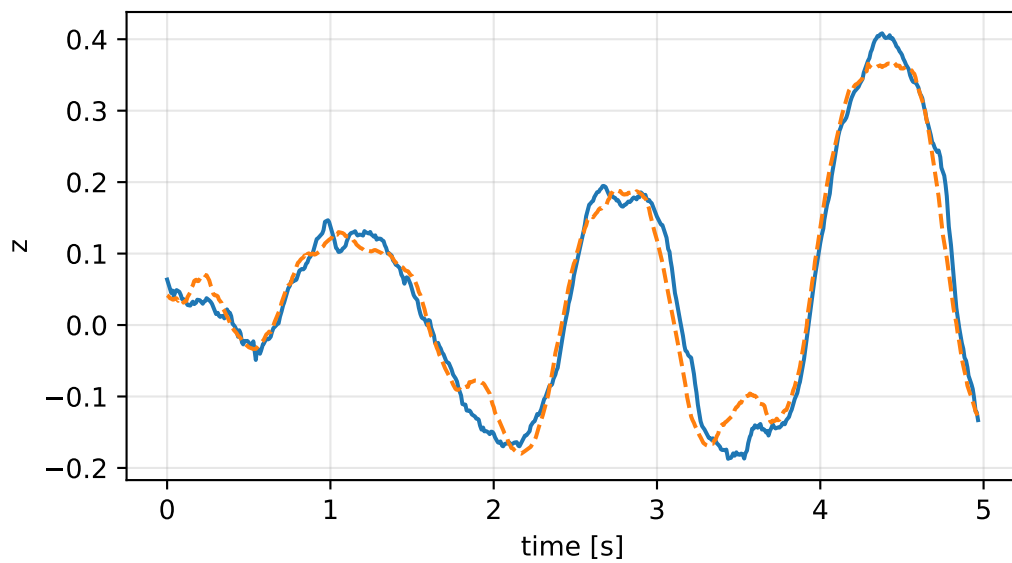
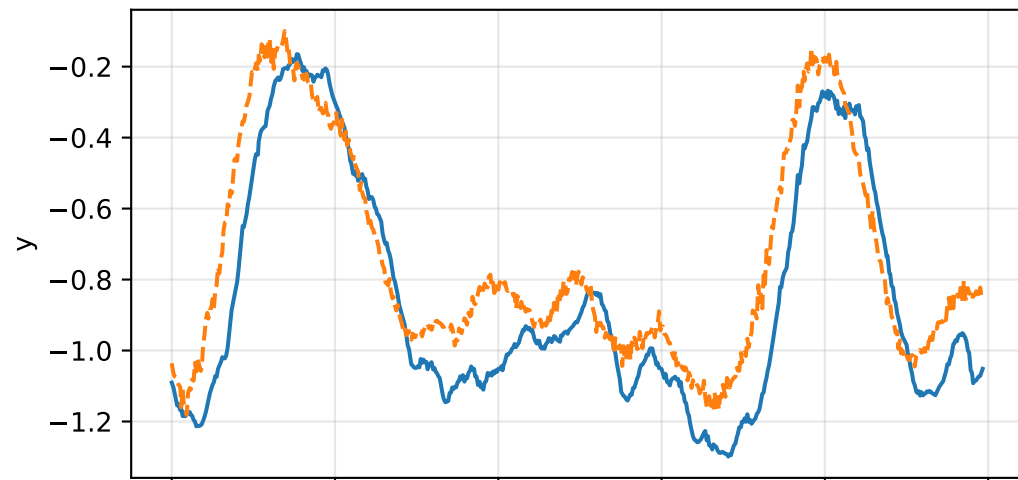
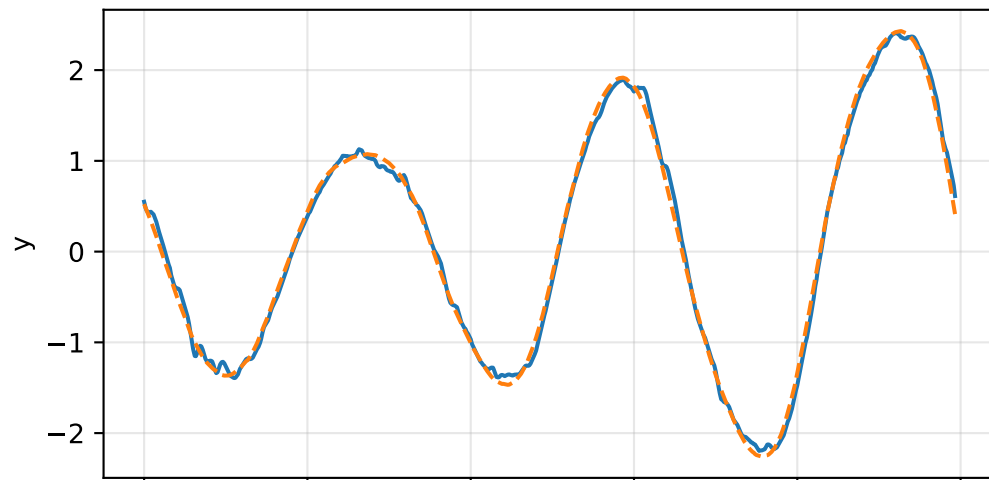
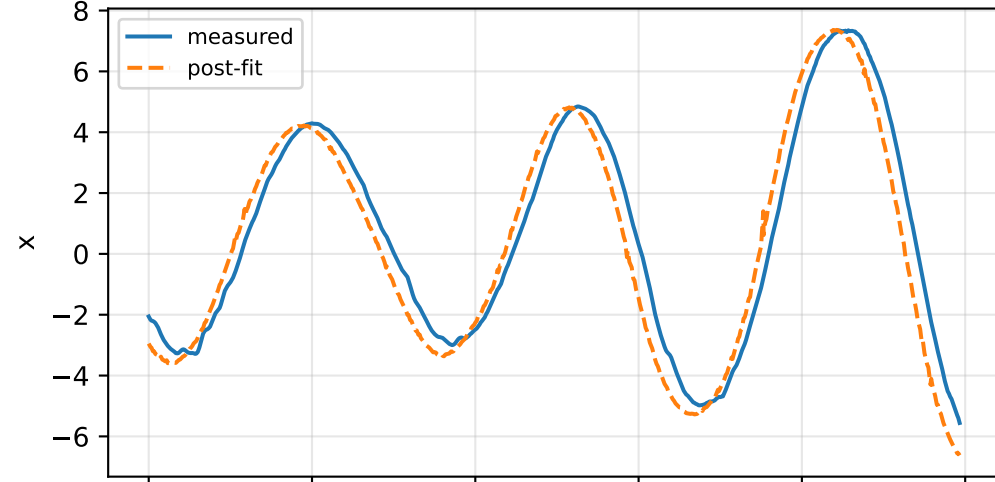


cog_prior_reseed_refined: IMU comparison (diagnostic only)

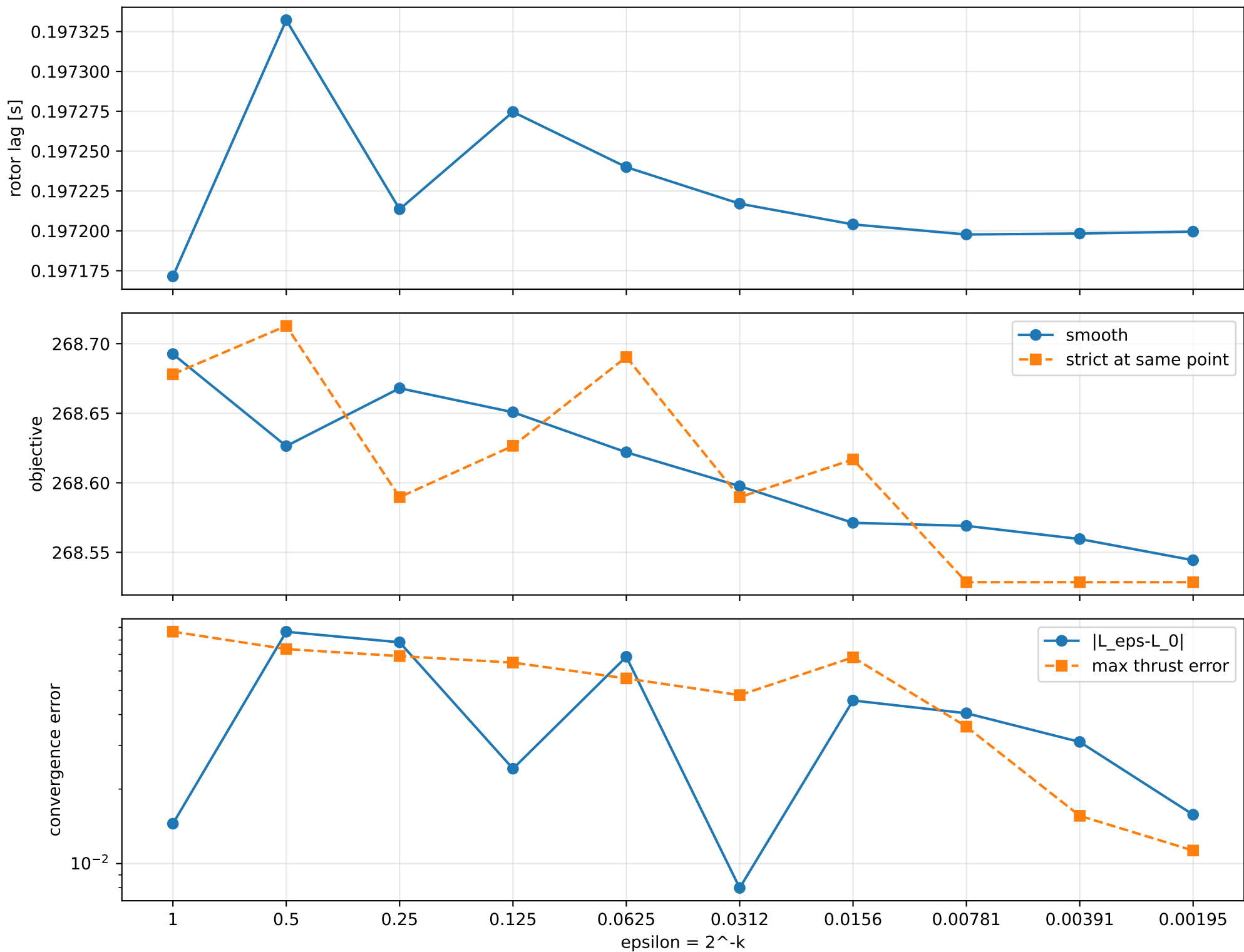
gyro [rad/s]



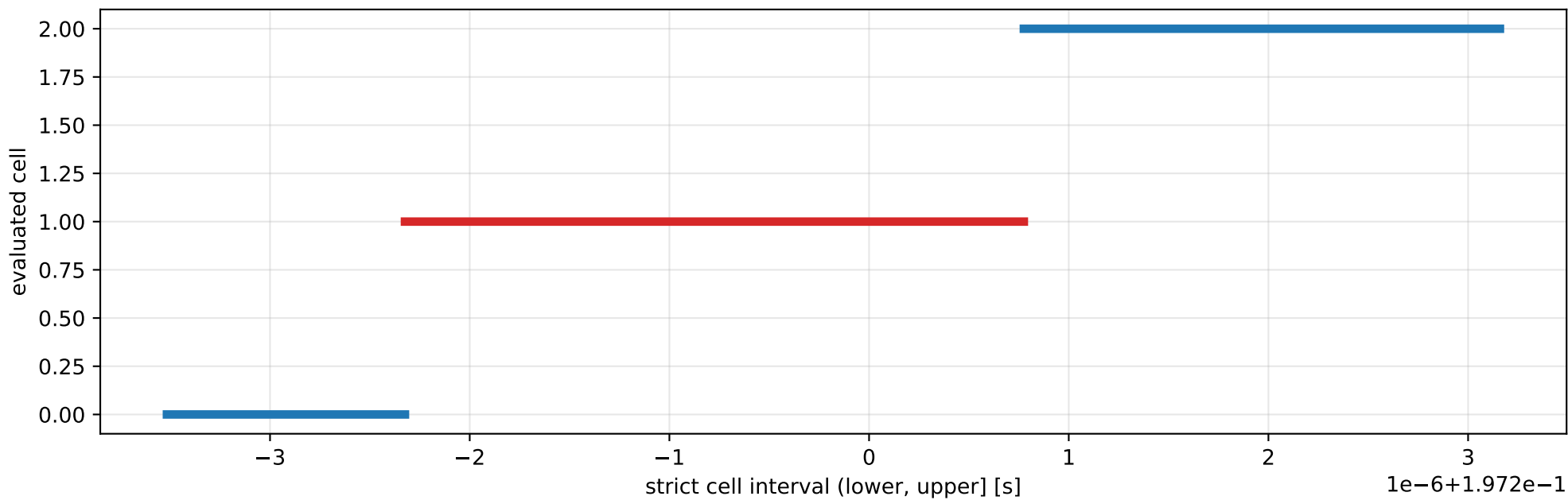
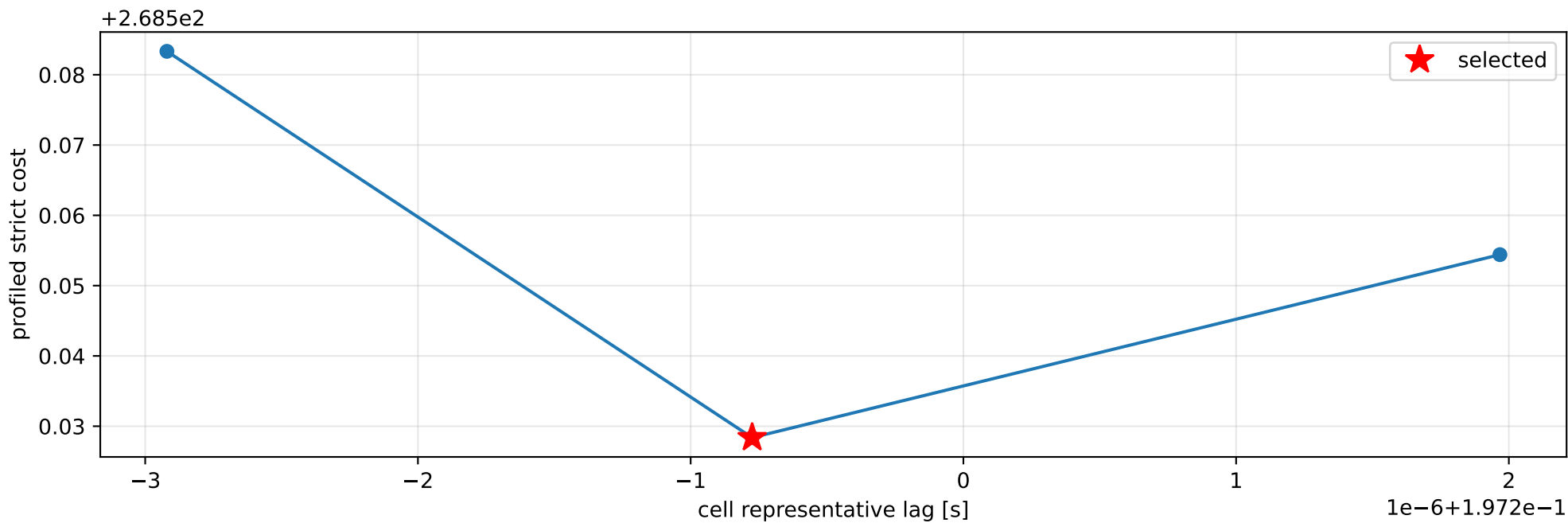
specific force [m/s^2]



cog_prior_reseed_refined: smooth-to-strict continuation



cog_prior_reseed_refined: exact strict-ZOH lag cells



selected (0.197197676, 0.197200775] s; final smooth 0.197199511 s; data boundary=False

cog_prior_reseed_refined: parameter estimate

Case: cog_prior_reseed_refined

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 4.66810473

inertia [kg m²]:

```
[[ 4.74214154e-01 -5.69699400e-04 -7.70305543e-02]
 [-5.69699400e-04  3.61103649e-01  4.62923433e-05]
 [-7.70305543e-02  4.62923433e-05  8.00802330e-01]]
```

CoG body [m]: [0.01762701 -0.00411798 -0.0443757]

force effectiveness: [1.56339933 2.01318079 1.75798389 1.45756899]

scale-free J/m [m²]:

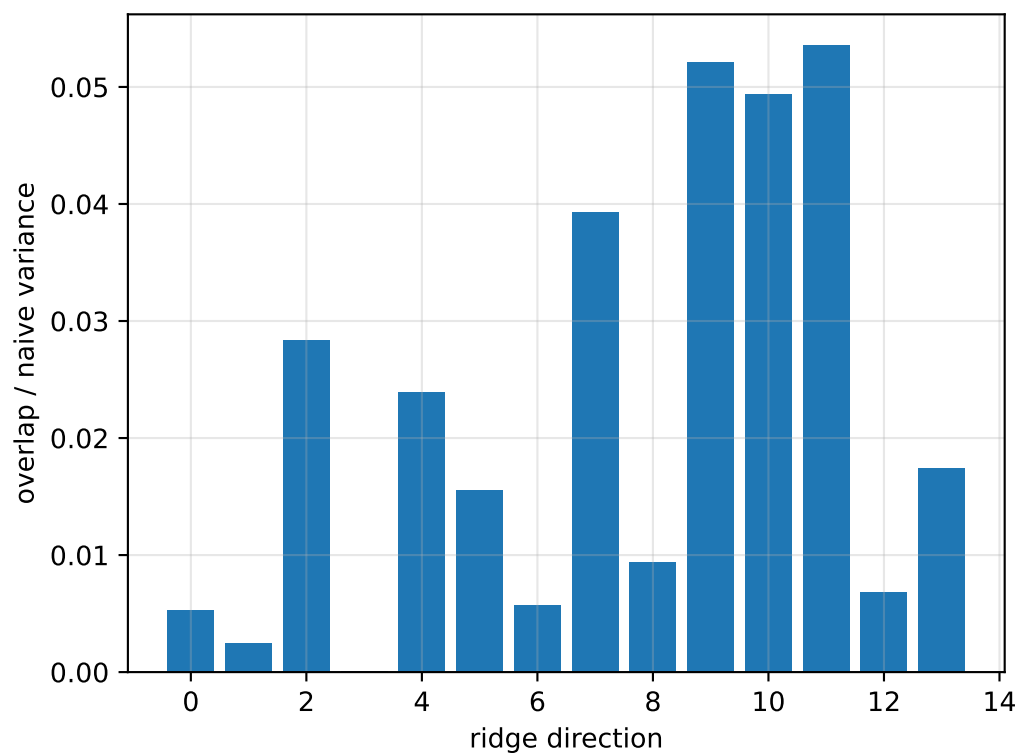
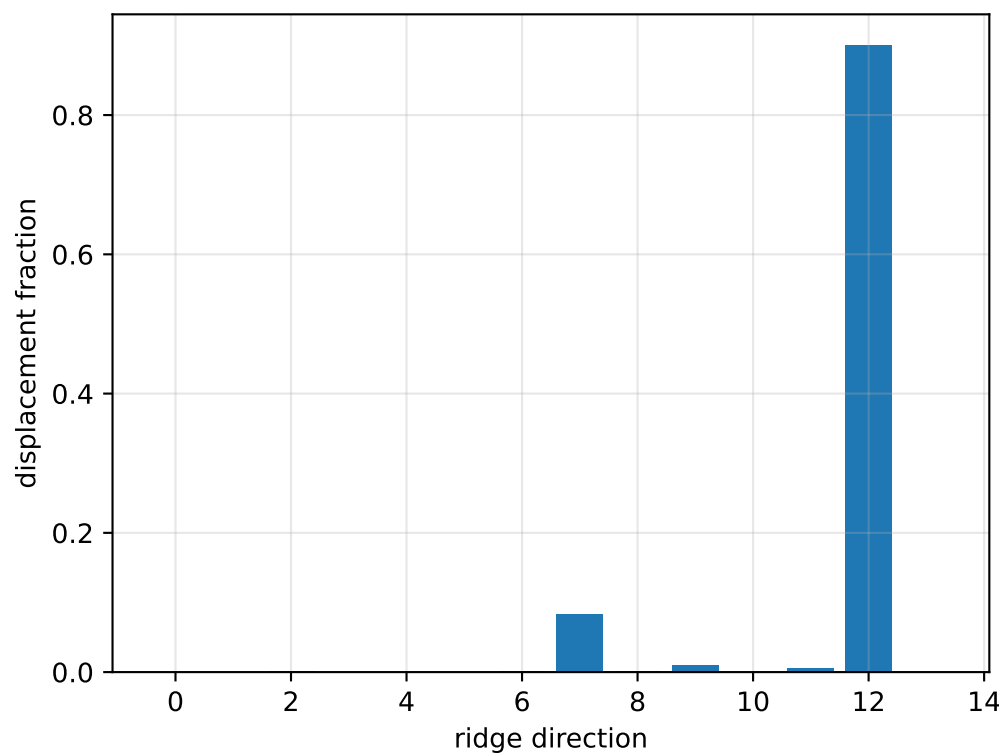
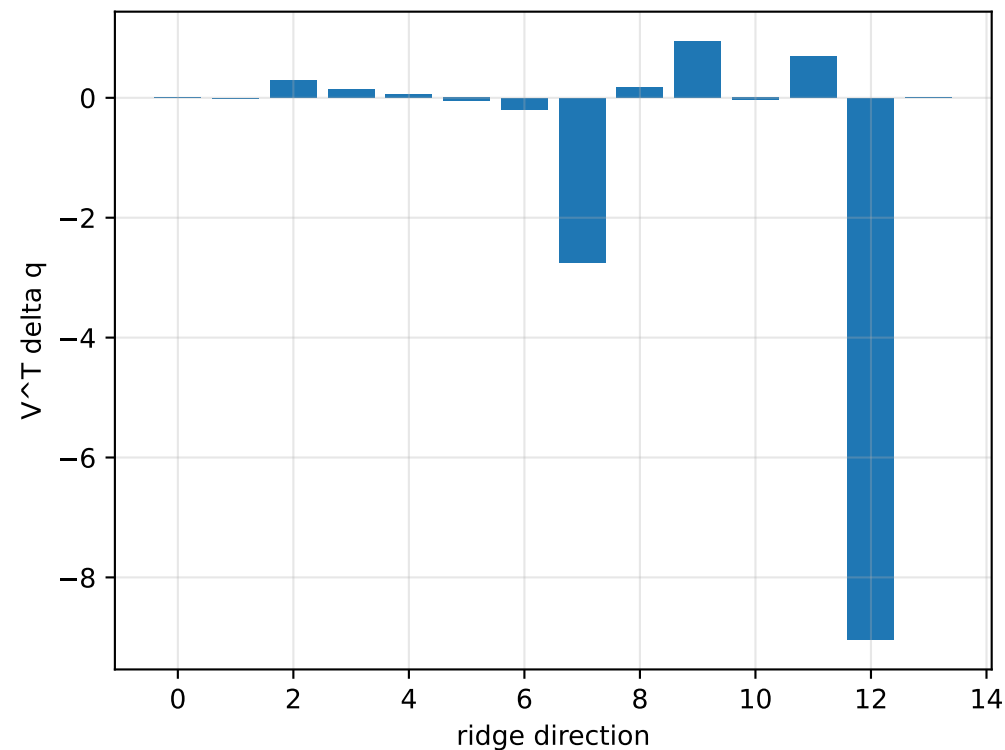
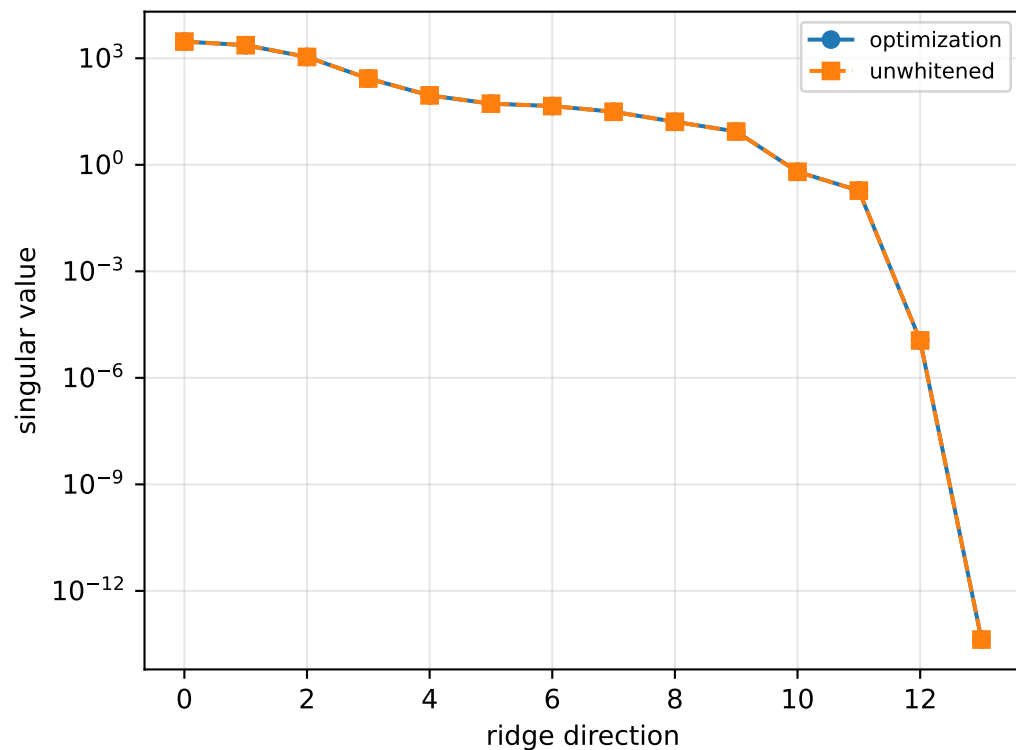
```
[[ 1.01586014e-01 -1.22040835e-04 -1.65014623e-02]
 [-1.22040835e-04  7.73555157e-02  9.91673195e-06]
 [-1.65014623e-02  9.91673195e-06  1.71547636e-01]]
```

scale-free f/m: [0.33491094 0.43126299 0.37659478 0.31223999]

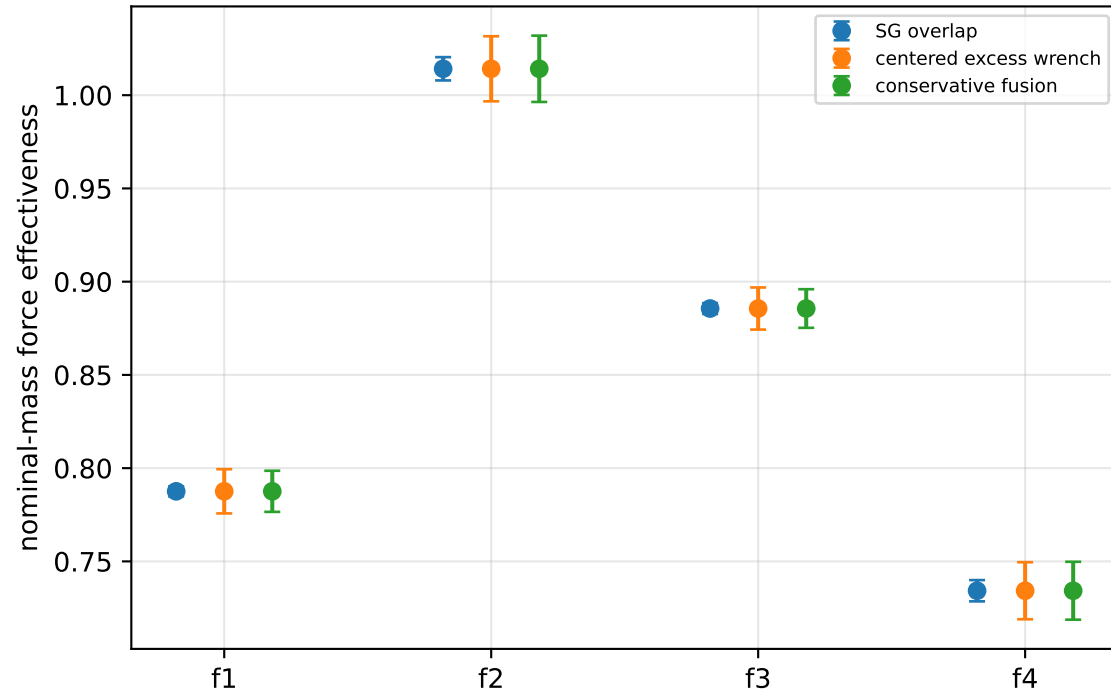
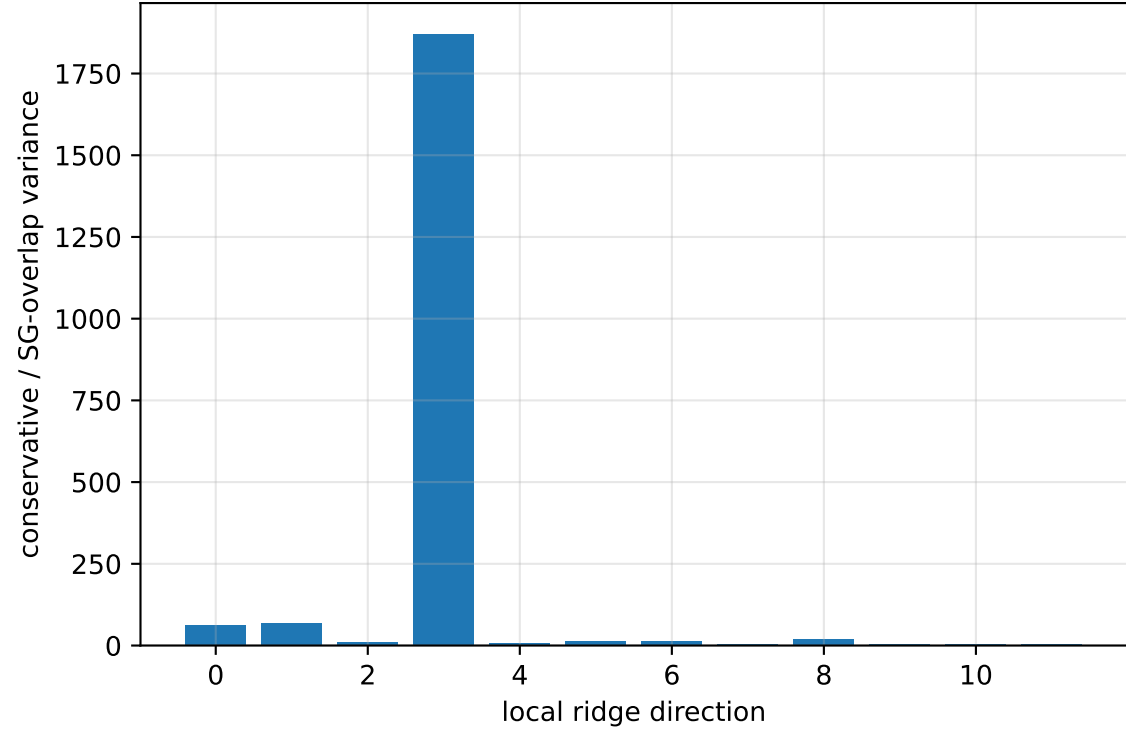
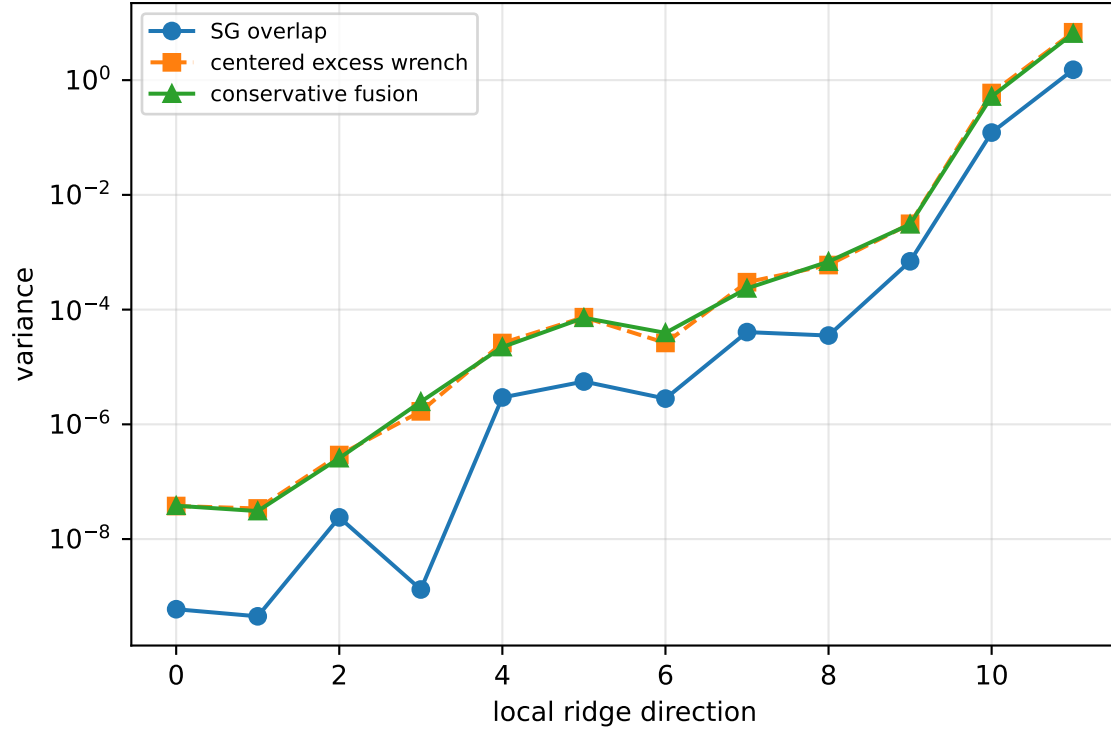
rotor lag cell: (0.197197676, 0.197200775] s

representative: 0.197199225 s

cog_prior_reseed_refined: ridge spectrum (rank 13, nullity 1, ||jv||=1.02e-13)



cog_prior_reseed_refined: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 44.21$
wrench-to-acceleration recovery max error = $2.8\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

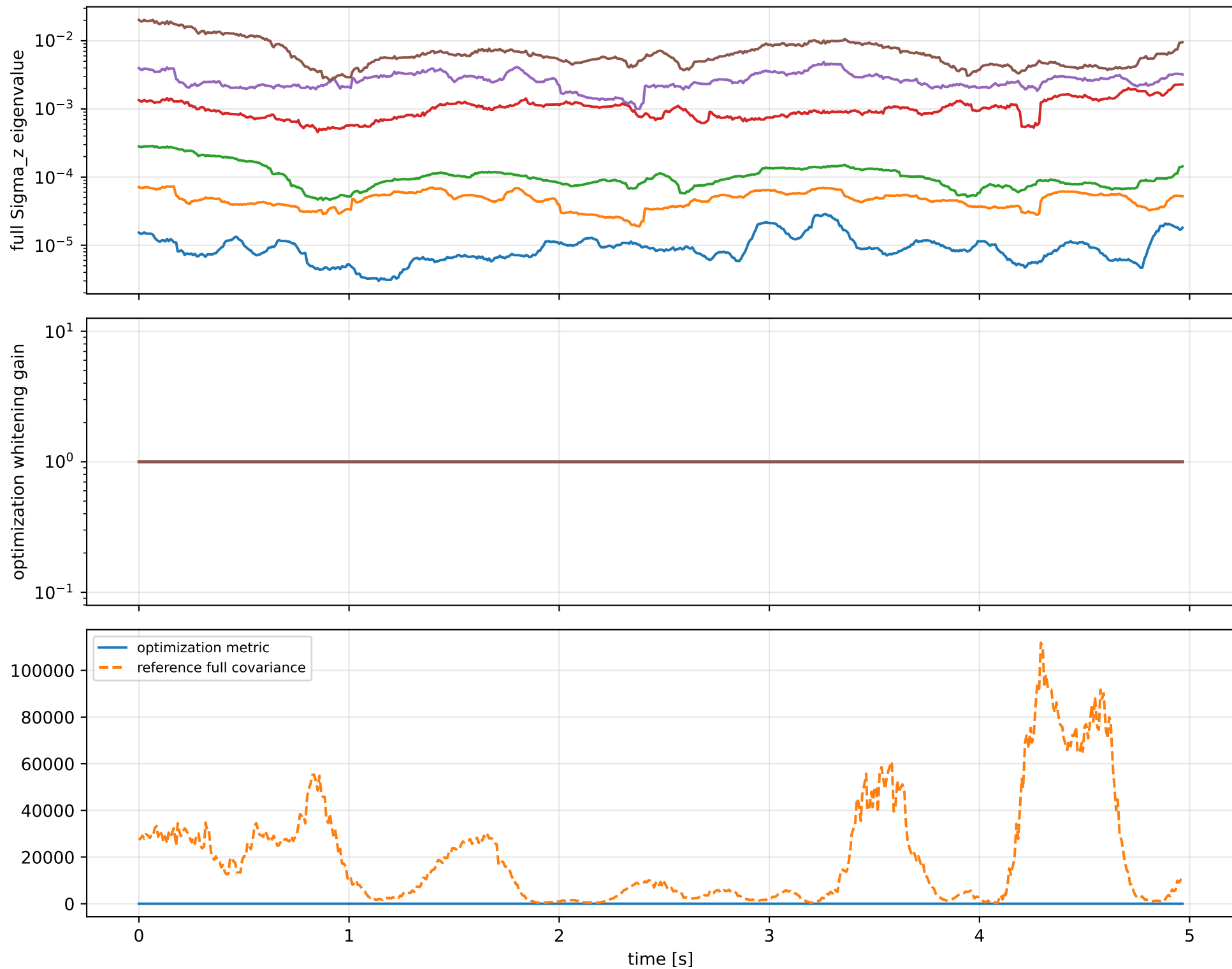
index 11: variance=6.484, ratio=4.272
 $\log_{\text{mass}} = -0.0395$; second_moment_diag_1=-0.428
second_moment_diag_2=-0.0382; second_moment_diag_3=+0.663
second_moment_offdiag_12=+0.0482; second_moment_offdiag_13=-0.604
second_moment_offdiag_23=-0.0263; cog_x=+6.15e-06
cog_y=-0.000163; cog_z=+1.76e-05
log_force_eff_1=-0.0391; log_force_eff_2=-0.0387
log_force_eff_3=-0.0403; log_force_eff_4=-0.0399

index 10: variance=0.5177, ratio=4.244
 $\log_{\text{mass}} = +0.00217$; second_moment_diag_1=+0.0242
second_moment_diag_2=+0.00804; second_moment_diag_3=-0.0398
second_moment_offdiag_12=-0.00642; second_moment_offdiag_13=+0.0289
second_moment_offdiag_23=-0.44; cog_x=+0.0011
cog_y=-0.00275; cog_z=+0.000262
log_force_eff_1=+0.00841; log_force_eff_2=+0.016
log_force_eff_3=-0.00374; log_force_eff_4=-0.0154

index 9: variance=0.003072, ratio=4.425
 $\log_{\text{mass}} = -0.147$; second_moment_diag_1=-0.0314
second_moment_diag_2=+0.899; second_moment_diag_3=-0.0885
second_moment_offdiag_12=-0.00642; second_moment_offdiag_13=-0.0816
second_moment_offdiag_23=+0.0117; cog_x=+0.0221
cog_y=-0.0137; cog_z=-0.0039
log_force_eff_1=-0.186; log_force_eff_2=-0.0213
log_force_eff_3=-0.0933; log_force_eff_4=-0.331

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

cog_prior_reseed_refined: optimization vs reference covariance



cog_prior_reseed_refined: wrench / closure (force max $1.32\text{e-}14$, torque max $9.02\text{e-}16$)

